

Research Statement

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Environmental quality depends on incentives. My research studies how changes in incentives reshape resource use, environmental quality, and social welfare. I focus on settings where private decisions generate environmental externalities that markets fail to internalize, spanning agriculture, forestry, transportation, and energy.

My work follows two complementary themes. The first examines how resource-dependent households and firms manage environmental resources as incentives change. The second studies how transportation systems generate externalities and how policies and shocks reshape travel behavior. Both themes ask how economic decisions interact with environmental constraints and public policy.

1. Environmental and Natural Resource Management

Agriculture and forestry rely on natural resources whose environmental value is rarely reflected in market prices. This research examines how resource-dependent households respond to changing incentives, and how those responses affect production, environmental outcomes, and risk.

- **Farm Household Economics** Liou et al. (2025a), Lee et al. (2025a), Chang and Liou (2026), and Lee et al. (2025b) examine how farm and forest households make production decisions, manage environmental resources, and bear economic risk. Renewable-energy adoption complements farming by raising farm sales and labor demand. Food-traceability adoption lowers chemical input use and groundwater extraction while improving revenue and market access. In forestry, greater tree diversity reduces expected income but also lowers income volatility, showing that biodiversity acts as natural capital with risk-reduction value.
- **Climate Policy** Liou et al. (2025b) extends the agenda to international climate policy. Using a theoretical model, it shows how carbon border adjustment policies alter incentives for cleaner production and reshape global production decisions.

2. Transportation and Environmental Externalities

Transportation generates large externalities through accidents, congestion, pollution, and carbon emissions. This research examines how transport policies and large-scale shocks affect travel behavior, environmental outcomes, and social welfare.

- **Large-scale Events and Road Safety** Liou and Chang (2026a), Liou and Chang (2026b), and Chang et al. (2024) study how institutional change and large-scale events affect traffic risk and environmental outcomes. Traffic-regulation reform reduces accidents, injuries, and medical costs, whereas large-scale events raise traffic risk and pollution mainly by shifting travel demand.
- **Travel Demand and COVID-19** Liou et al. (2025c), Chang et al. (2021), Liou et al. (2024), and Liou et al. (2026) examine how transport systems and local economies respond to policy changes and large-scale shocks. Transit pricing raises mobility while cutting carbon emissions, and COVID-19 reshaped travel behavior and slowed economic recovery by changing perceived risk and consumer decisions.

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